

Description

The cybernetic system is a two-way Morse Code translator. In Text-to-Morse mode, the microcontroller receives text input from the connected laptop via UART. The program then encodes the text to Morse code and plays the morse by blinking an RGB LED, buzzing a connected active buzzer, and displaying the Morse code on the onboard LCD upon pressing the onboard user button. The mode can be switched by a connected external button. In Morse-to-Text mode, an external python script running on the connected laptop uses computer vision to interpret specific hand signals as Morse code and translates it to English. The microcontroller then receives the translated text via UART and displays it on the onboard LCD upon pressing the onboard user button.

Requirements

- STM32F429-Discovery
- Mini-USB
- Breadboard
- Breadboard wires
- Active buzzer
- RGB LED
- 470 Ω resistor

Finite State Machine (FSM)

Table 1: Description of Each State

State 0	State 1	State 2	State 3
Display idle screen for Text-to-Morse mode	Display and play the translated morse	Display the translated text	Display idle screen for Morse-to-Text mode

Table 2: State Transition Table for FSM

Inputs		States	
User Button	External Button	Current	Next
0	0	0	0
		1	0
		2	2
		3	2
0	1	0	2
		1	2
		2	0
		3	0
1	0	0	1

		1	1
		2	3
		3	3

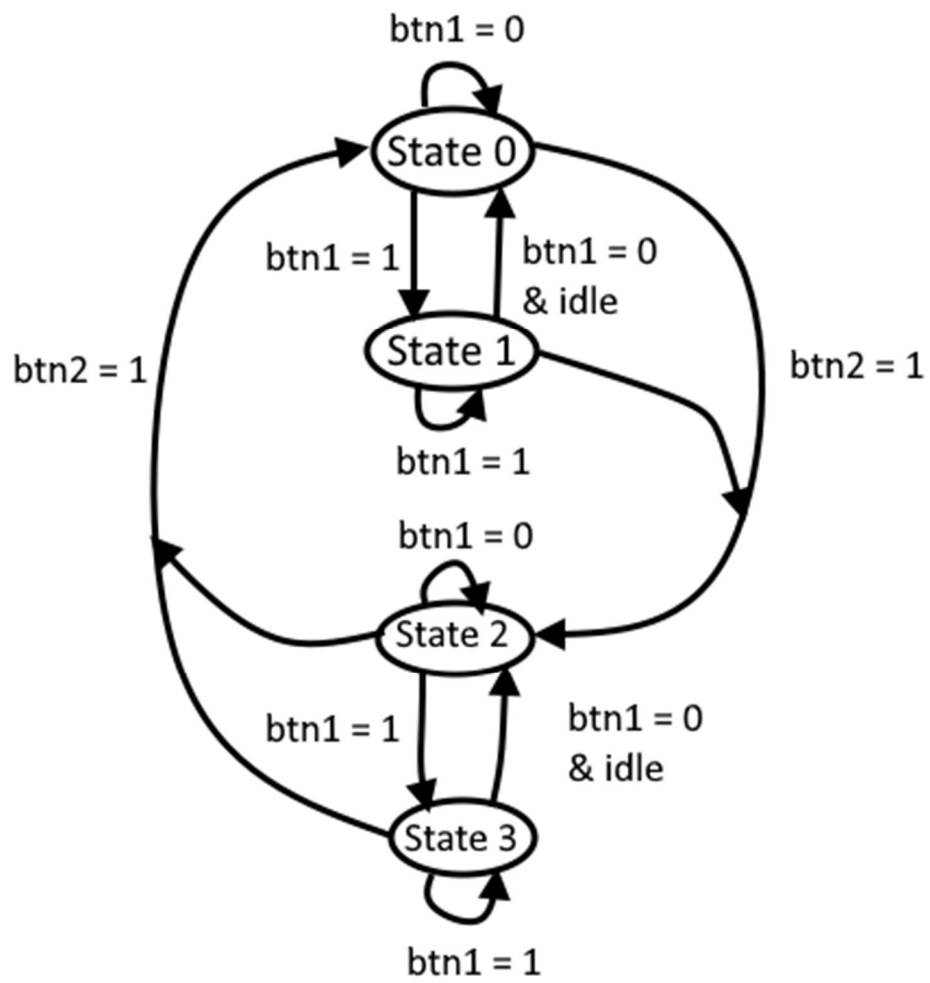


Figure 1: Diagram of FSM

Documentation

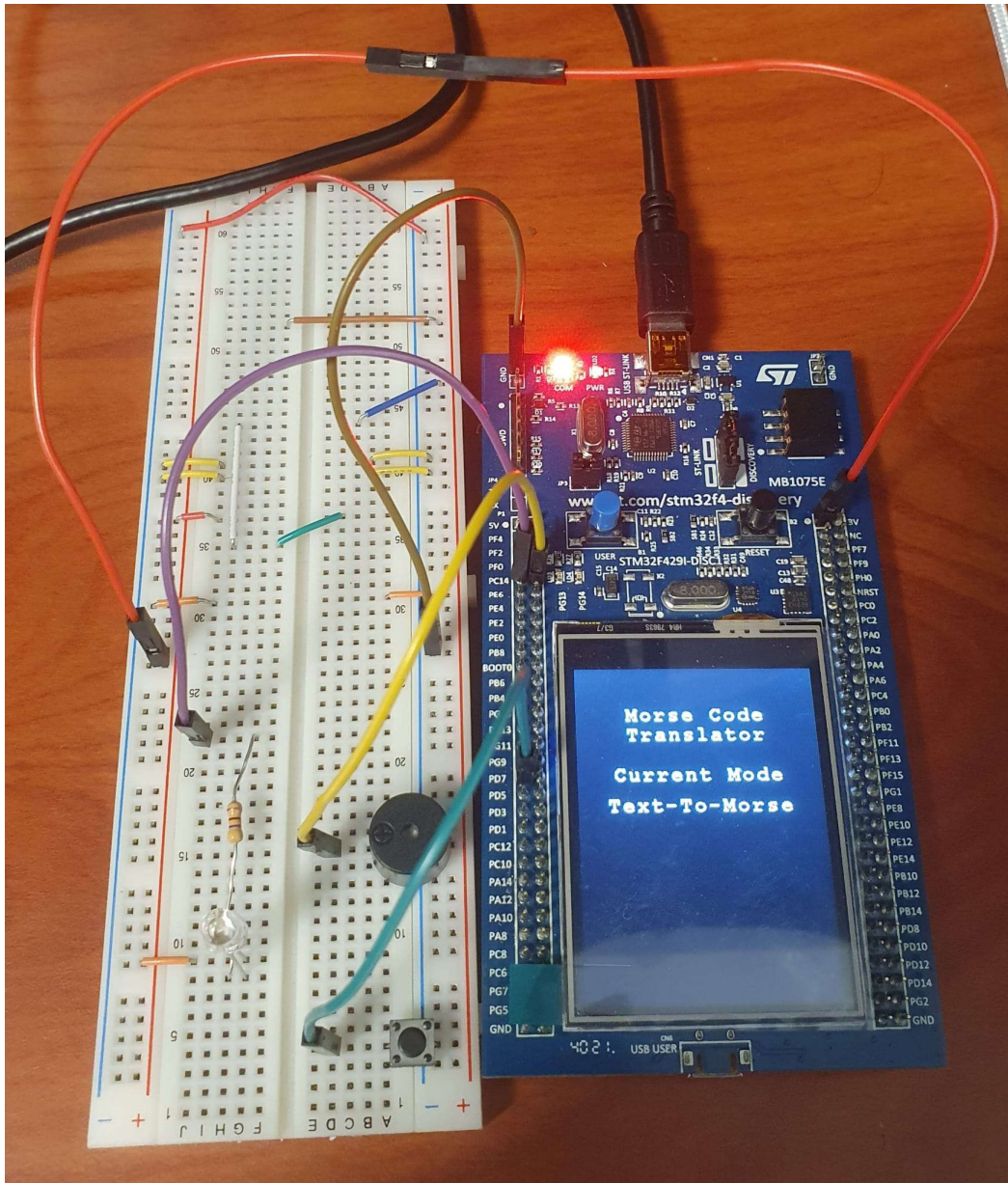


Figure 2: Cybernetic System in Text-to-Morse Idle State